MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE WDU

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389 except as modified in 3.4.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.06 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389 with the exception that the operating temperature range shall be 0° to 80°C.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be WDU.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/21-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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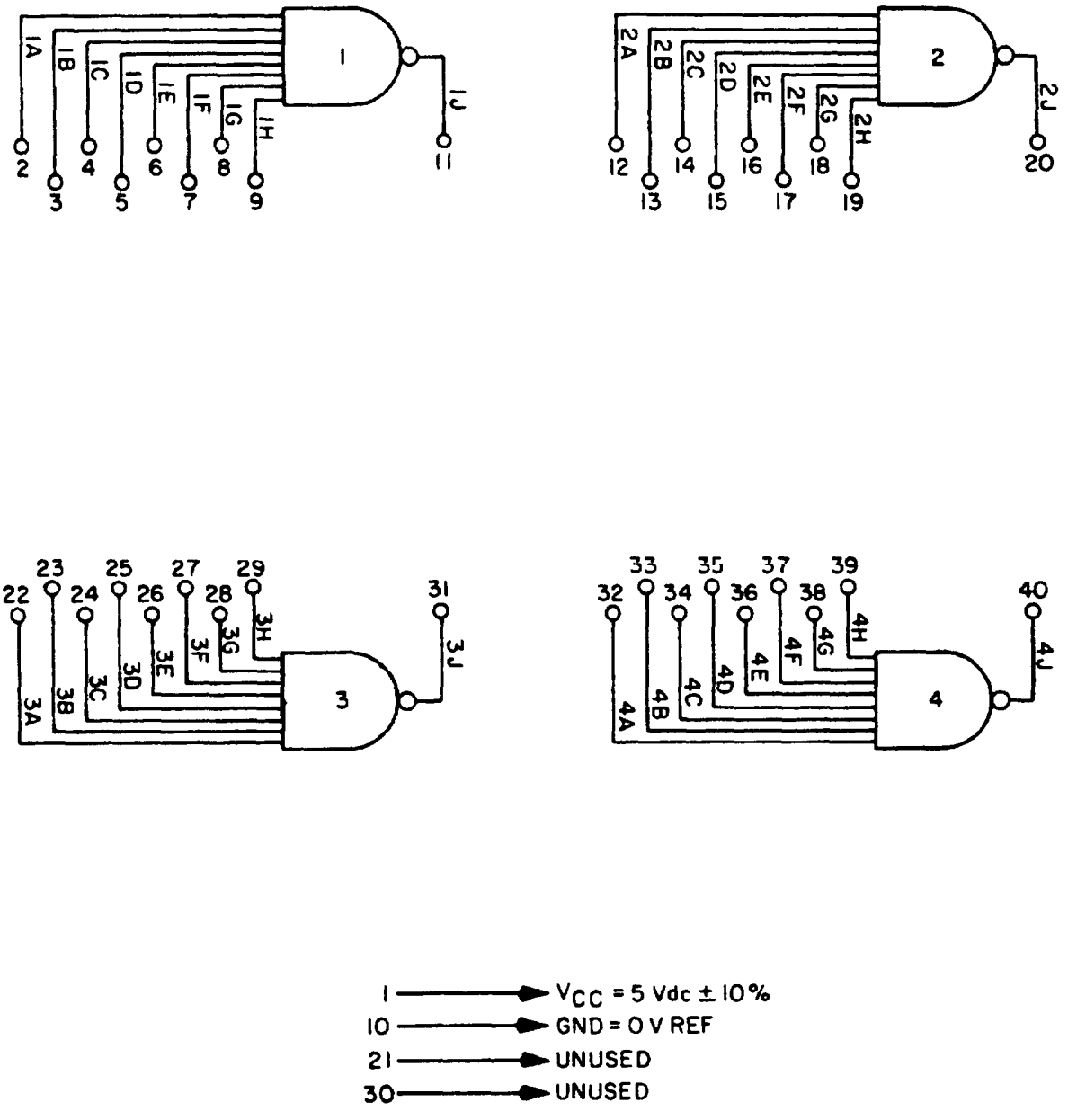


FIGURE 1. WDU module function diagram, NAND gate .

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-B32	Static input current	IV	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	mA
C1-C32	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D4	Output voltage, low level	VI	X	0	450	0	450	0	450	0	450	mV
E1-E32	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F32	Input current, low level	VII	X	0.1	2.2	0.1	2.2	0.1	2.2	0.1	2.2	mA
G1-G224	Input current, high level	VII	X	0	60	0	60	0	60	0	60	µA
H1-H4	Output short circuit current	VII	X	40	100	35	100	35	100	25	100	mA
J1	Power supply current, low level	VIII	S	10	40	10	40	10	40	10	40	mA
J2	Power supply current, high level	VIII	S	5.0	16.8	5.0	16.8	5.0	16.8	5.0	16.8	mA
K1-K4	Transitiopn time, t_{PHL}	IX	S	3	20	3	20	3	25	3	25	ns
L1-L4	Transitiopn time, t_{PLH}	IX	S	3	20	3	20	3	25	3	25	ns
M1-M4	Propagation delay time, t_{PLH}	IX	S	3	20	3	20	3	25	3	25	ns
N1-N4	Propagation delay time, t_{PHL}	IX	S	3	20	3	20	3	25	3	25	ns
P	Power supply transient amplitude	X	Q	0	400	0	400	0	400	0	400	mV
Q	Isolation	XI	S	10	∞	10	∞	10	∞	10	∞	MΩ
R1-R4	Output crosstalk, low level	XII	Q	0	400	0	400	0	400	0	400	mV
S1-S4	Output crosstalk, high level	XII	Q	0	400	0	400	0	400	0	400	mV
T1-T32	Input capacitance	XIII	Q	2	12	2	12	2	12	2	12	pF

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U	Power supply transient rise time	XIV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or Q are part of a sequence of tests, the order of testing shall be as follows: A, B, Q where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N and U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/21 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658888. Part number M28787/21-1 supersedes part numbers 2658888 and M28787/21.

6.4 Module description. The following electrical parameters reflect the 0° to 80°C end-of-life limits from table II.

Supply voltage	5 Vdc ±10 percent
Maximum power dissipation	
All outputs at low level	220 mW (outputs not loaded)
All outputs at high level	93 mW (outputs not loaded)

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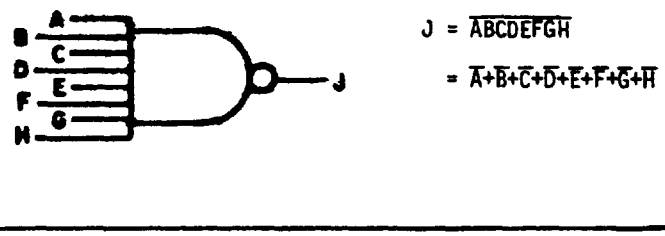
Input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	2.2 mA maximum
High level (V_{IH} = 2.4 V)	60 µA maximum
Output voltage	
Low level	0.45 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output drive	
Low level (V_{OL} = 0 V)	22 mA maximum
High level (V_{OH} = 2.4 V)	0.6 mA minimum

6.5 Module function. The module contains four 8-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input	Output
Any input at low level	High level
All inputs at high level	Low level



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

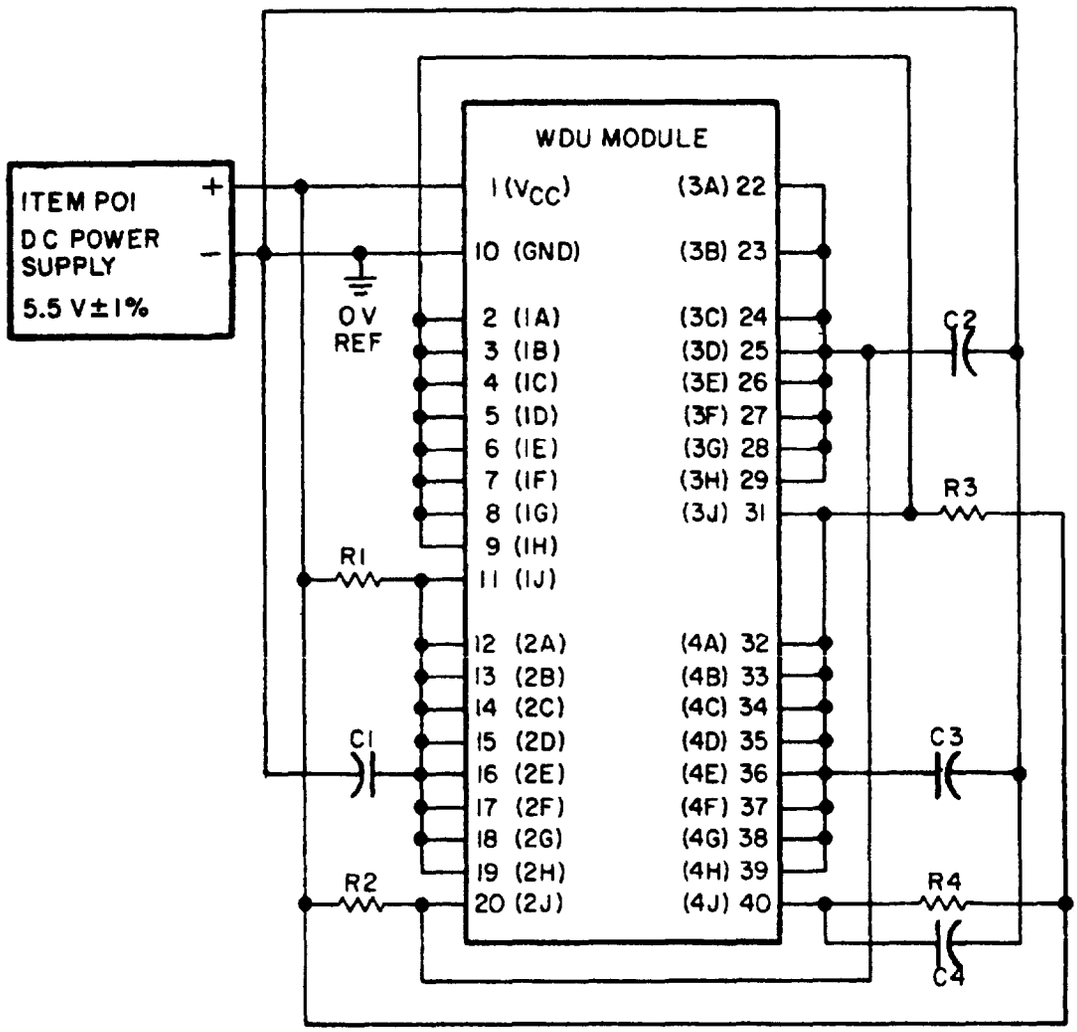
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cuationed to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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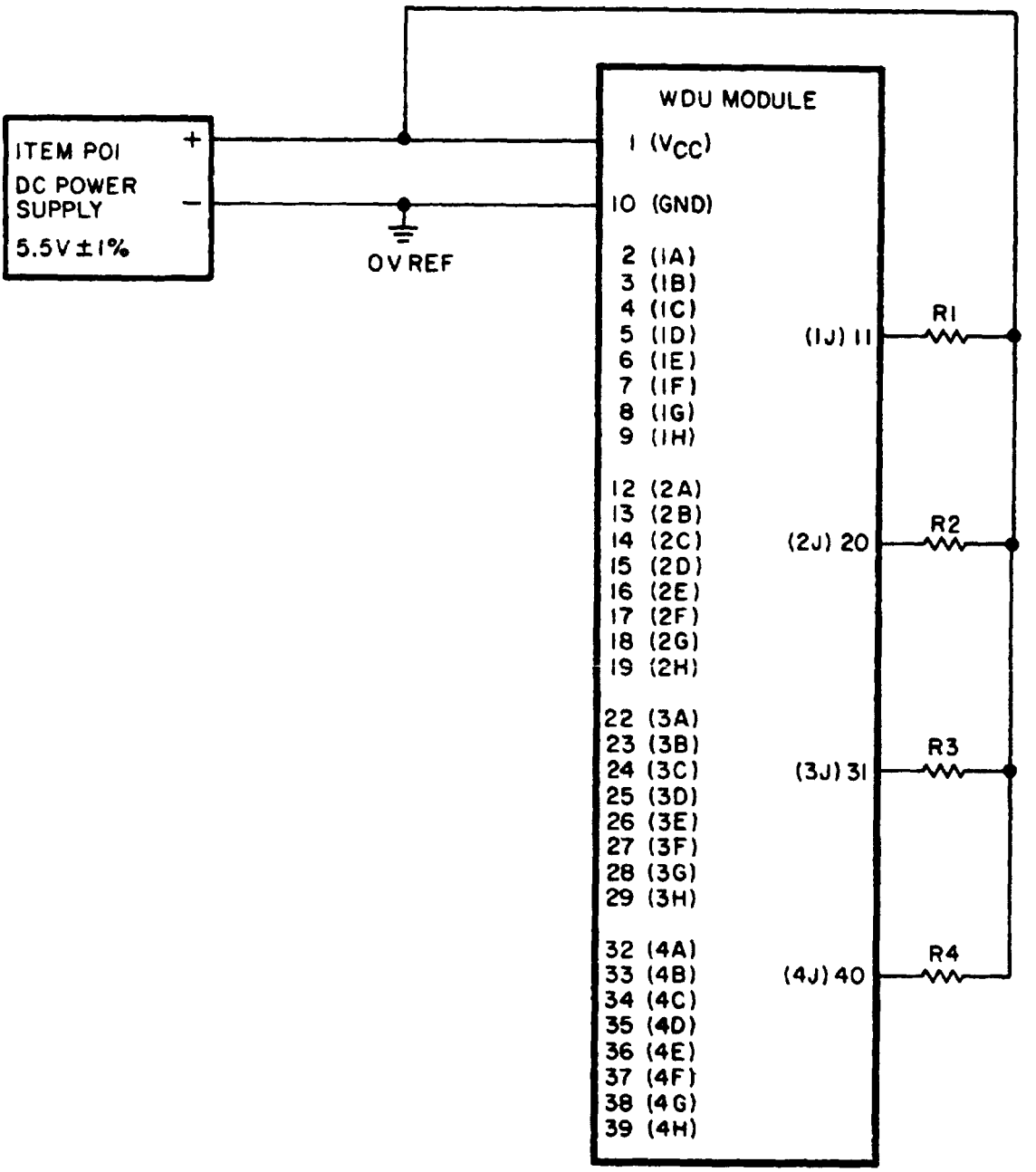
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NOTES
1. R1-R4 = 243Ω ±1%, 1/4 W.
2 C1-C4 = 130 pF ±20%.

FIGURE 2 Operating temperature and life test circuit.

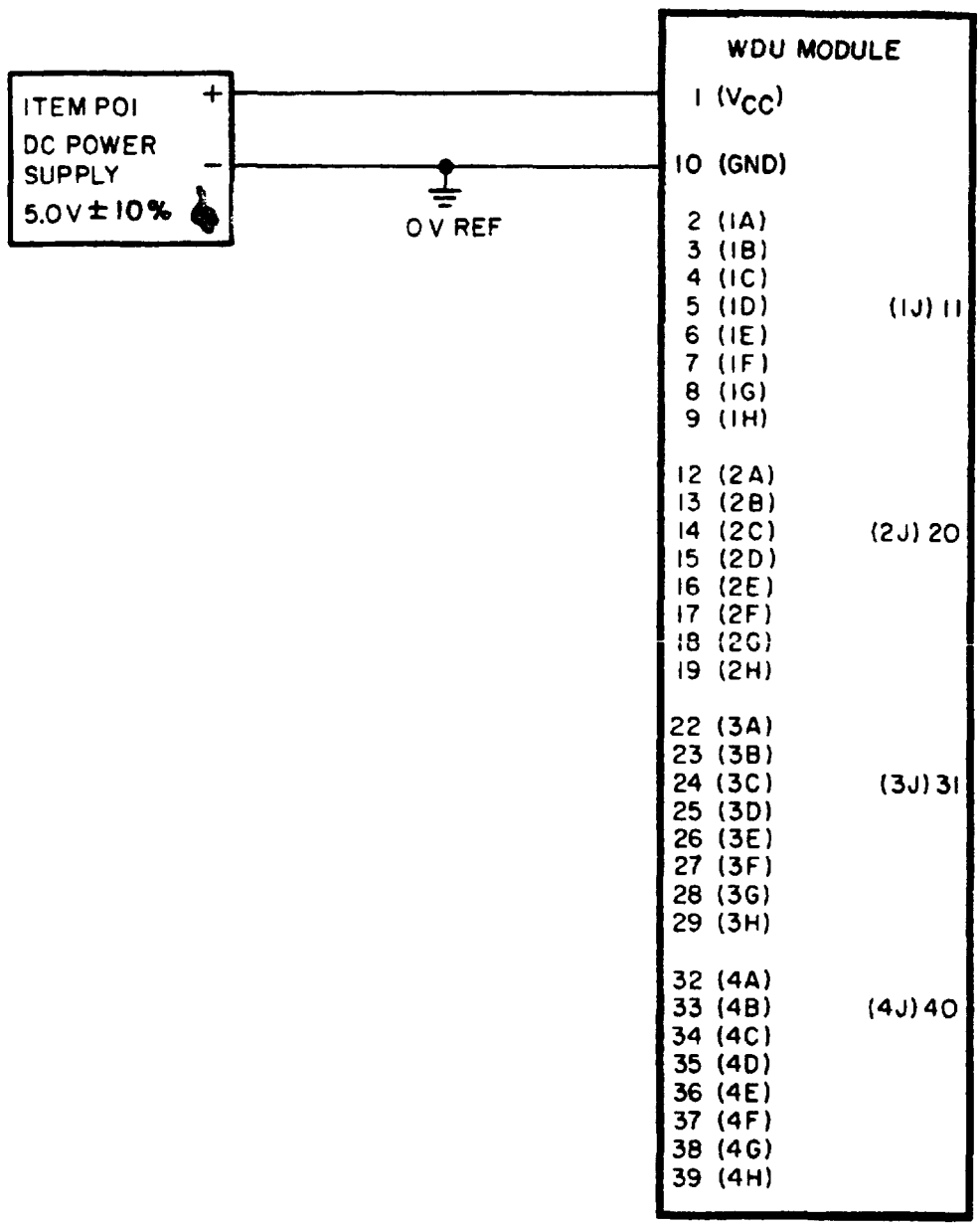
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NOTE: R1-R4 = 243Ω ±1%, 1/4 W.

FIGURE 3. Transient thermal test circuit.

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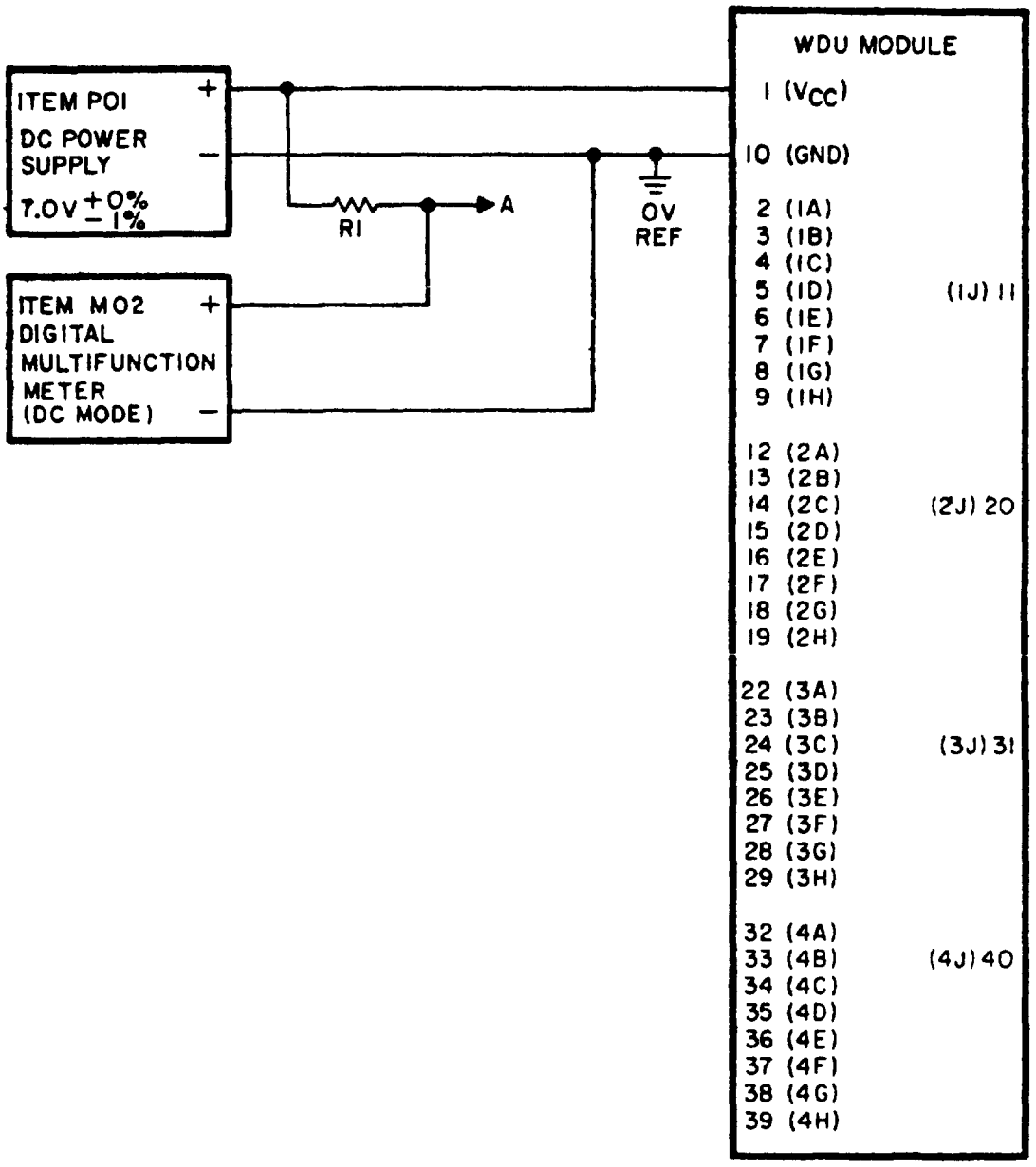
* FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1J V_{OUT} (low loaded)	11	0.1	0.8	V	5
A2	2J	20	↓	↓	↓	↓
A3	3J	31				
A4	4J	40				

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NOTES:
1. RI = 243Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Static input current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Static input current	2	3,4,5,6,7,8,9	0.0	0.5	mA	6
B2	1B	3	2,4,5,6,7,8,9				
B3	1C	4	2,3,5,6,7,8,9				
B4	1D	5	2,3,4,6,7,8,9				
B5	1E	6	2,3,4,5,7,8,9				
B6	1F	7	2,3,4,5,6,8,9				
B7	1G	8	2,3,4,5,6,7,9				
B8	1H	9	2,3,4,5,6,7,8				
B9	2A	12	13,14,15,16,17,18,19				
B10	2B	13	12,14,15,16,17,18,19				
B11	2C	14	12,13,15,16,17,18,19				
B12	2D	15	12,13,14,16,17,18,19				
B13	2E	16	12,13,14,15,17,18,19				
B14	2F	17	12,13,14,15,16,18,19				
B15	2G	18	12,13,14,15,16,17,19				
B16	2H	19	12,13,14,15,16,17,18				
B17	3A	22	23,24,25,26,27,28,29				
B18	3B	23	22,24,25,26,27,28,29				
B19	3C	24	22,23,25,26,27,28,29				
B20	3D	25	22,23,24,26,27,28,29				

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TABLE IV. Static input current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B21	3E	Static input current	26	22,23,24,25,27,28,29	0.0	0.5	mA	6
B22	3F	↓	27	22,23,24,25,26,28,29	↓	↓	↓	↓
B23	3G		28	22,23,24,25,26,27,29				
B24	3H		29	22,23,24,25,26,27,28				
B25	4A		32	33,34,35,36,37,38,39				
B26	4B		33	32,34,35,36,37,38,39				
B27	4C		34	32,33,35,36,37,38,39				
B28	4D		35	32,33,34,36,37,38,39				
B29	4E		36	32,33,34,35,37,38,39				
B30	4F		37	32,33,34,35,36,38,39				
B31	4G		38	32,33,34,35,36,37,39				
B32	4H		39	32,33,34,35,36,37,38				

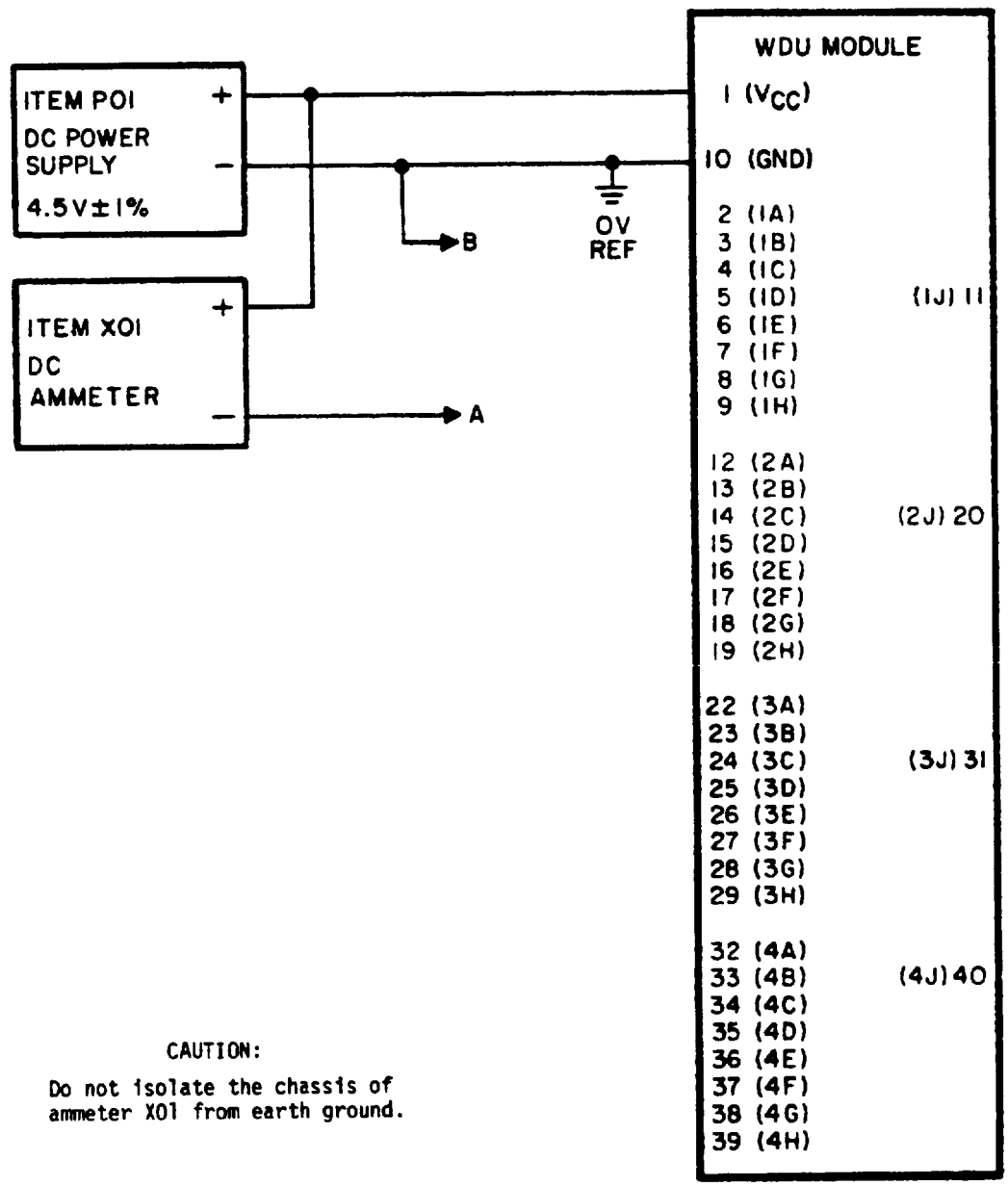


FIGURE 6. Static input current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

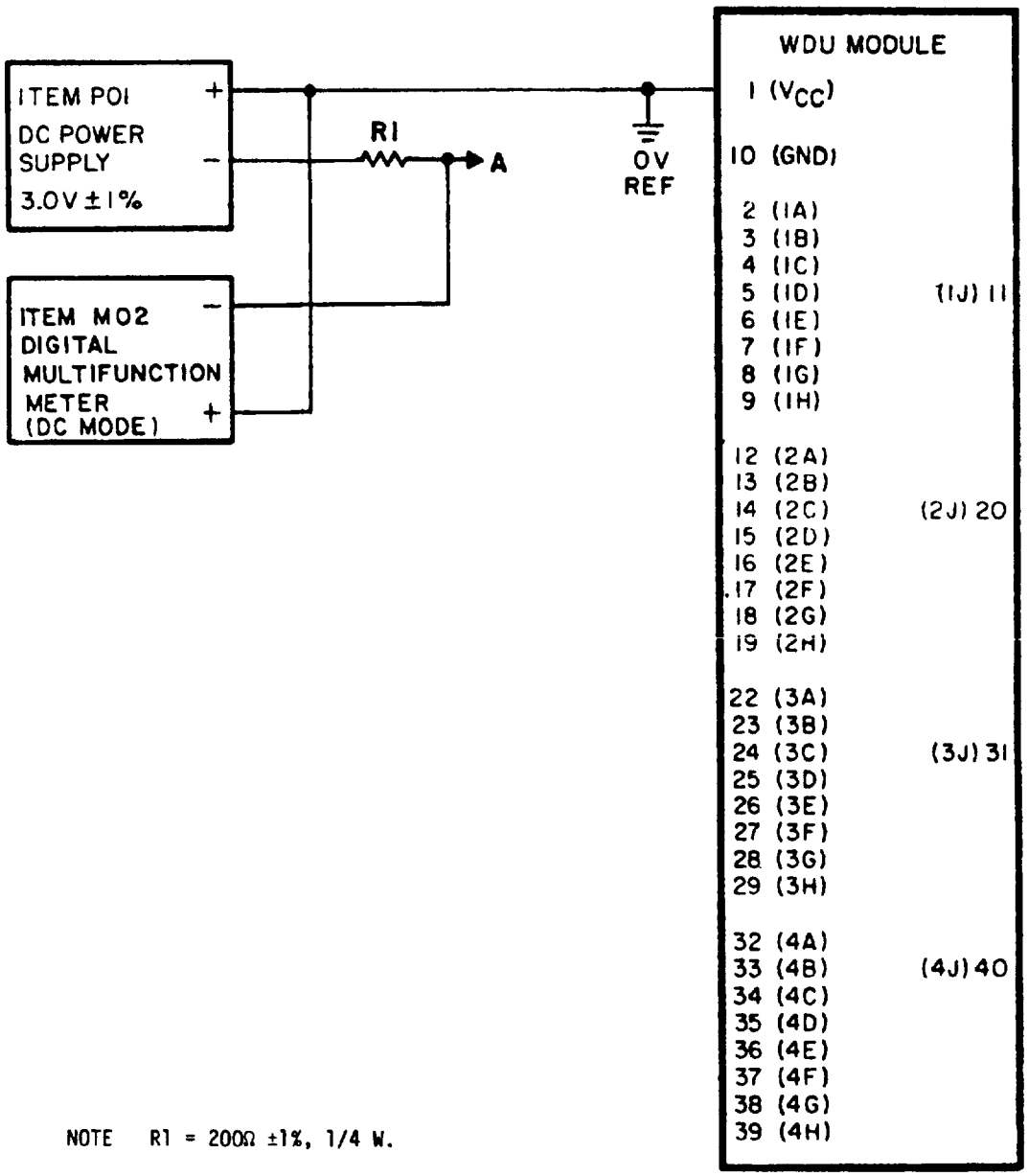
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1A Input clamp diode voltage	2	-0.4	-1.5	V	7
C2	1B	3				
C3	1C	4				
C4	1D	5				
C5	1E	6				
C6	1F	7				
C7	1G	8				
C8	1H	9				
C9	2A	12				
C10	2B	13				
C11	2C	14				
C12	2D	15				
C13	2E	16				
C14	2F	17				
C15	2G	18				
C16	2H	19				
C17	3A	22				
C18	3B	23				
C19	3C	24				
C20	3D	25				
C21	3E	26				

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TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C22	3F Input clamp diode voltage	27	-0.4	-1.5	V	7
C23	3G	28				
C24	3H	29				
C25	4A	32				
C26	4B	33				
C27	4C	34				
C28	4D	35				
C29	4E	36				
C30	4F	37				
C31	4G	38				
C32	4H	39				



NOTE R1 = 200Ω ±1%, 1/4 W.

* FIGURE 7. Input clamp diode voltage test circuit for table V.

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* TABLE VI. Output voltage (low-level and high-level) test.

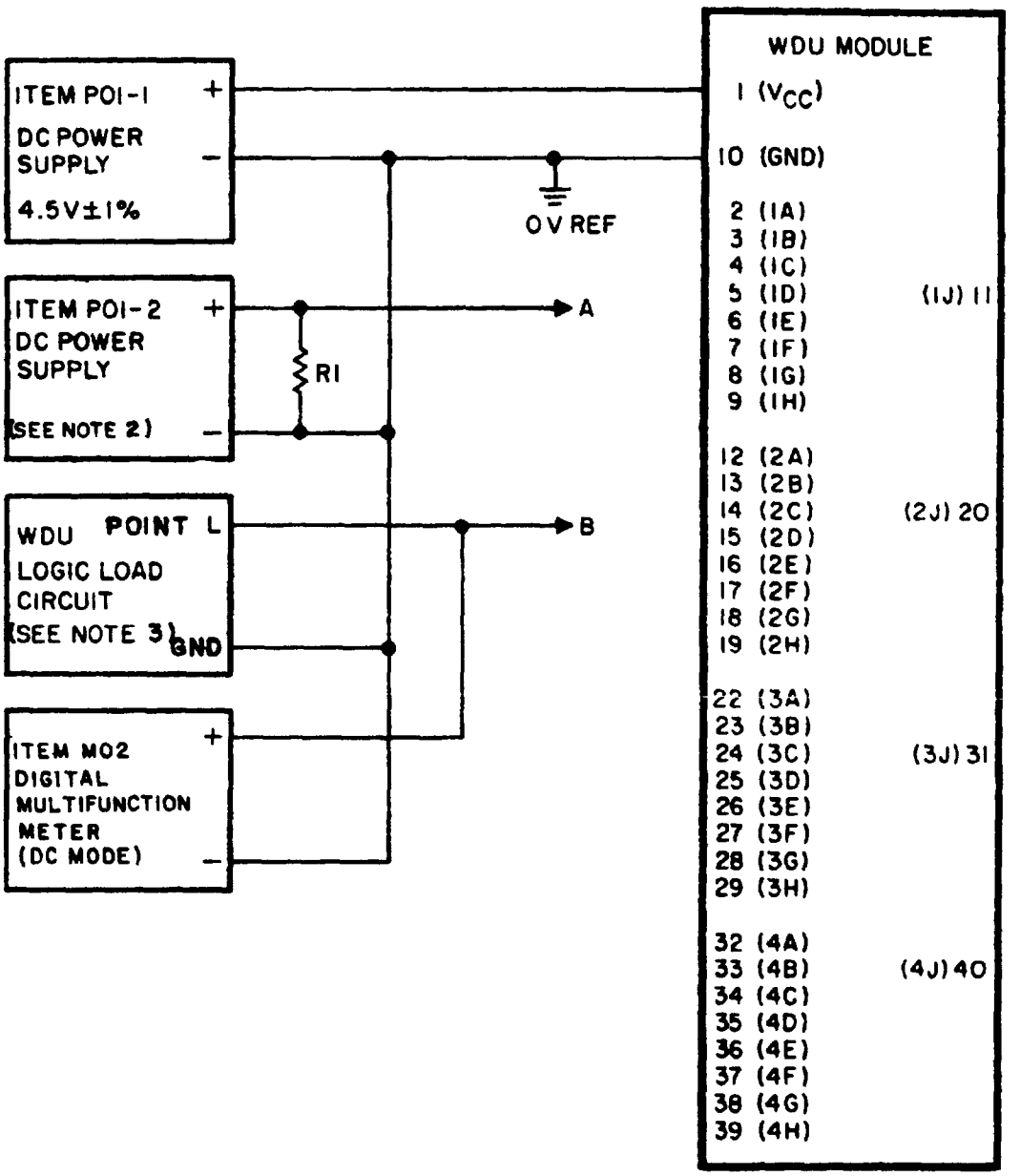
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1J Output voltage, low level	2,3,4,5,6,7,8.9	11	0	450	mV	8
D2	2J	12,13,14,15,16,17, 18,19	20				
D3	3J	22,23,24,25,26,27, 28,29	31				
D4	4J	32,33,34,35,36,37, 38,39	40				
E1	1A Output voltage, high level	2	11	2.4	4.5	V	
E2	1B	3					
E3	1C	4					
E4	1D	5					
E5	1E	6					
E6	1F	7					
E7	1G	8					
E8	1H	9					
E9	2A	12	20				
E10	2B	13					
E11	2C	14					
E12	2D	15					
E13	2E	16					
E14	2F	17					
E15	2G	18					

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* TABLE VI. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E16	2H Output voltage, high level	19	20	2.4	4.5	V	8
E17	3A	22	31				
E18	3B	23					
E19	3C	24					
E20	3D	25					
E21	3E	26					
E22	3F	27					
E23	3G	28					
E24	3H	29					
E25	4A	32	40				
E26	4B	33					
E27	4C	34					
E28	4D	35					
E29	4E	36					
E30	4F	37					
E31	4G	38					
E32	4H	39					

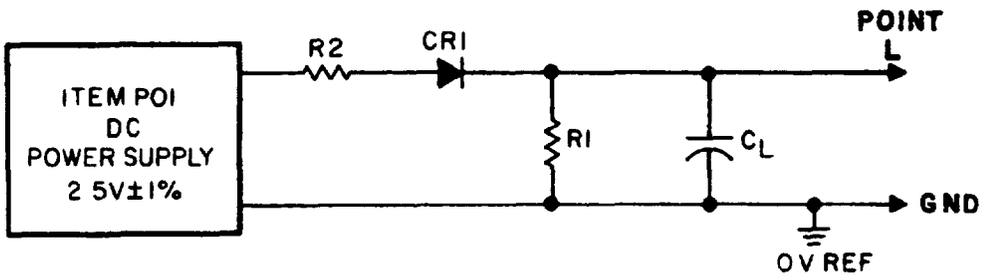
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NOTES:
1. R1 = 49.9Ω ±1%, 1 W.
2. Set item PO1-2 voltage as follows:
   V_{IN} = 2.0 V ±1% for test conditions D1-D4.
   V_{IN} = 0.8 V ±1% for test conditions E1-E32.
3. Load each output with a load as specified on figure 9.

* FIGURE 8. Static output voltage (low-level and high-level) test circuit for table VI.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance).
3. R1 = 4.02 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 22 mA ±1% from point L to ground.
5. Capacitor C_L is not required for test conditions D1-D4 and E1-E32
6. During output switching, item PO1 shall have a maximum overshoot of 500 mV.

FIGURE 9. WDU logic load circuit.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1A	Input current, low level	3,4,5,6,7,8,9	2	Open	0.1	2.2	mA	10
F2	1B	↓	2,4,5,6,7,8,9	3	↓	↓	↓	↓	↓
F3	1C		2,3,5,6,7,8,9	4					
F4	1D		2,3,4,6,7,8,9	5					
F5	1E		2,3,4,5,7,8,9	6					
F6	1F		2,3,4,5,6,8,9	7					
F7	1G		2,3,4,5,6,7,9	8					
F8	1H		2,3,4,5,6,7,8	9					
F9	2A		13,14,15,16,17,18,19	12					
F10	2B		12,14,15,16,17,18,19	13					
F11	2C		12,13,15,16,17,18,19	14					
F12	2D		12,13,14,16,17,18,19	15					
F13	2E		12,13,14,15,17,18,19	16					
F14	2F		12,13,14,15,16,18,19	17					
F15	2G		12,13,14,15,16,17,19	18					
F16	2H		12,13,14,15,16,17,18	19					
F17	3A		23,24,25,26,27,28,29	22					
F18	3B		22,24,25,26,27,28,29	23					
F19	3C		22,23,25,26,27,28,29	24					
F20	3D		22,23,24,26,27,28,29	25					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F21	3E	Input current, low level	22,23,24,25,27,28,29	26	Open	0.1	2.2	mA	10
F22	3F		22,23,24,25,26,28,29	27					
F23	3G		22,23,24,25,26,27,29	28					
F24	3H		22,23,24,25,26,27,28	29					
F25	4A		33,34,35,36,37,38,39	32					
F26	4B		32,34,35,36,37,38,39	33					
F27	4C		32,33,35,36,37,38,39	34					
F28	4D		32,33,34,36,37,38,39	35					
F29	4E		32,33,34,35,37,38,39	36					
F30	4F		32,33,34,35,36,38,39	37					
F31	4G		32,33,34,35,36,37,39	38					
F32	4H		32,33,34,35,36,37,38	39					
G1	1A	Input current, high level	2	Open	3	0	60	µA	
G2	1A				4				
G3	1A				5				
G4	1A				6				
G5	1A				7				
G6	1A				8				
G7	1A				9				
G8	1B		3		2				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G9	1B	Input curret, high level	3	Open	4	0	60	µA	10
G10	1B				5				
G11	1B				6				
G12	1B				7				
G13	1B				8				
G14	1B				9				
G15	1C		4		2				
G16	1C				3				
G17	1C				5				
G18	1C				6				
G19	1C				7				
G20	1C				8				
G21	1C				9				
G22	1D		5		2				
G23	1D				3				
G24	1D				4				
G25	1D				6				
G26	1D				7				
G27	1D				8				
G28	1D				9				
G29	1E		6		2				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G30	1E Input current, high level	6	Open	3	0	60	µA	10
G31	1E			4				
G32	1E			5				
G33	1E			7				
G34	1E			8				
G35	1E			9				
G36	1F	7		2				
G37	1F			3				
G38	1F			4				
G39	1F			5				
G40	1F			6				
G41	1F			8				
G42	1F			9				
G43	1G	8		2				
G44	1G			3				
G45	1G			4				
G46	1G			5				
G47	1G			6				
G48	1G			7				
G49	1G			9				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G50	1H Input current, high level	9	Open	2	0	60	µA	10
G51	1H			3				
G52	1H			4				
G53	1H			5				
G54	1H			6				
G55	1H			7				
G56	1H			8				
G57	2A	12		13				
G58	2A			14				
G59	2A			15				
G60	2A			16				
G61	2A			17				
G62	2A			18				
G63	2A			19				
G64	2B	13		12				
G65	2B			14				
G66	2B			15				
G67	2B			16				
G68	2B			17				
G69	2B			18				
G70	2B			19				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G71	2C Input current, high level	14	Open	12	0	60	µA	10
G72	2C			13				
G73	2C			15				
G74	2C			16				
G75	2C			17				
G76	2C			18				
G77	2C			19				
G78	2D	15		12				
G79	2D			13				
G80	2D			14				
G81	2D			16				
G82	2D			17				
G83	2D			18				
G84	2D			19				
G85	2E	16		12				
G86	2E			13				
G87	2E			14				
G88	2E			15				
G89	2E			17				
G90	2E			18				
G91	2E			19				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G92	2F	Input current, high level	17	Open	12	0	60	µA	10
G93	2F				13				
G94	2F				14				
G95	2F				15				
G96	2F				16				
G97	2F				18				
G98	2F				19				
G99	2G		18		12				
G100	2G				13				
G101	2G				14				
G102	2G				15				
G103	2G				16				
G104	2G				17				
G105	2G				19				
G106	2H		19		12				
G107	2H				13				
G108	2H				14				
G109	2H				15				
G110	2H				16				
G111	2H				17				
G112	2H				18				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G113	3A Input current, high level	22	Open	23	0	60	µA	10
G114	3A			24				
G115	3A			25				
G116	3A			26				
G117	3A			27				
G118	3A			28				
G119	3A			29				
G120	3B	23		22				
G121	3B			24				
G122	3B			25				
G123	3B			26				
G124	3B			27				
G125	3B			28				
G126	3B			29				
G127	3C	24		22				
G128	3C			23				
G129	3C			25				
G130	3C			26				
G131	3C			27				
G132	3C			28				
G133	3C			29				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G134	3D	Input current, high level	25	Open	22	0	60	µA	10
G135	3D				23				
G136	3D				24				
G137	3D				26				
G138	3D				27				
G139	3D				28				
G140	3D				29				
G141	3E		26		22				
G142	3E				23				
G143	3E				24				
G144	3E				25				
G145	3E				27				
G146	3E				28				
G147	3E				29				
G148	3F		27		22				
G149	3F				23				
G150	3F				24				
G151	3F				25				
G152	3F				26				
G153	3F				28				
G154	3F				29				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G155	3G Input current, high level	28	Open	22	0	60	µA	10
G156	3G			23				
G157	3G			24				
G158	3G			25				
G159	3G			26				
G160	3G			27				
G161	3G			29				
G162	3H	29		22				
G163	3H			23				
G164	3H			24				
G165	3H			25				
G166	3H			26				
G167	3H			27				
G168	3H			28				
G169	4A	32		33				
G170	4A			34				
G171	4A			35				
G172	4A			36				
G173	4A			37				
G174	4A			38				
G175	4A			39				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G176	4B Input current, high level	33	Open	32	0	60	µA	10
G177	4B			34				
G178	4B			35				
G179	4B			36				
G180	4B			37				
G181	4B			38				
G182	4B			39				
G183	4C	34		32				
G184	4C			33				
G185	4C			35				
G186	4C			36				
G187	4C			37				
G188	4C			38				
G189	4C			39				
G190	4D	35		32				
G191	4D			33				
G192	4D			34				
G193	4D			36				
G194	4D			37				
G195	4D			38				
G196	4D			39				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

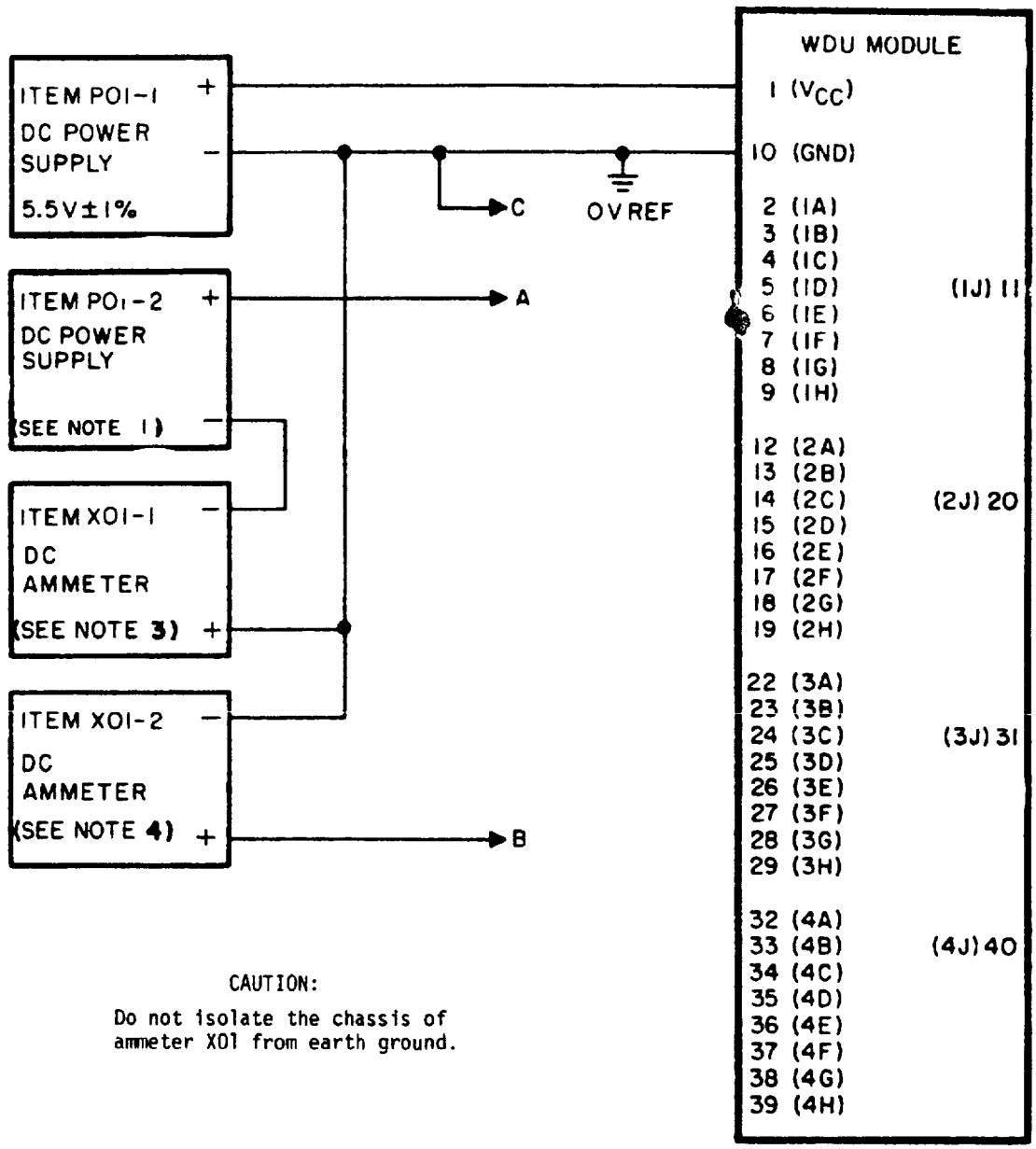
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G197	4E Input current, high level	36	Open	32	0	60	µA	10
G198	4E			33				
G199	4E			34				
G200	4E			35				
G201	4E			37				
G202	4E			38				
G203	4E			39				
G204	4F	37		32				
G205	4F			33				
G206	4F			34				
G207	4F			35				
G208	4F			36				
G209	4F			38				
G210	4F			39				
G211	4G	38		32				
G212	4G			33				
G213	4G			34				
G214	4G			35				
G215	4G			36				
G216	4G			37				
G217	4G			39				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G218	4H	Input current, high level	39	Open	32	0	60	µA	10
G219	4H				33				
G220	4H				34				
G221	4H				35				
G222	4H				36				
G223	4H				37				
G224	4H				38				
H1	1J	Output short circuit current	Open	11	2,3,4,5, 6,7,8,9	40	100	mA	
H2	2J			20	12,13,14, 15,16,17, 18,19				
H3	3J			31	22,23,24, 25,26,27, 28,29				
H4	4J			40	32,33,34, 35,36,37, 38,39				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Set item PO1-2 voltage as follows:
   V_{IN} = 4.5 V ±1% for test conditions F1-F32.
   V_{IN} = 2.4 V ±1% for test conditions G1-G224.
   V_{IN} = 0 (A is open) for test conditions H1-H4.
2. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
3. Use item XO1-1 for test conditions G1-G224. With point A disconnected, item XO1-1 should indicate 0 current on the 1 µA range.
4. Use item XO1-2 for test conditions F1-F32 and H1-H4.
5. When performing the output short circuit current test, only one output shall be shorted at a time and the measurement taken within 5 seconds.

✱ FIGURE 10 Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
J1	Power supply current, low level	8	10	40	mA	11
J2	Power supply current, high level	10	5.0	16.8	mA	11

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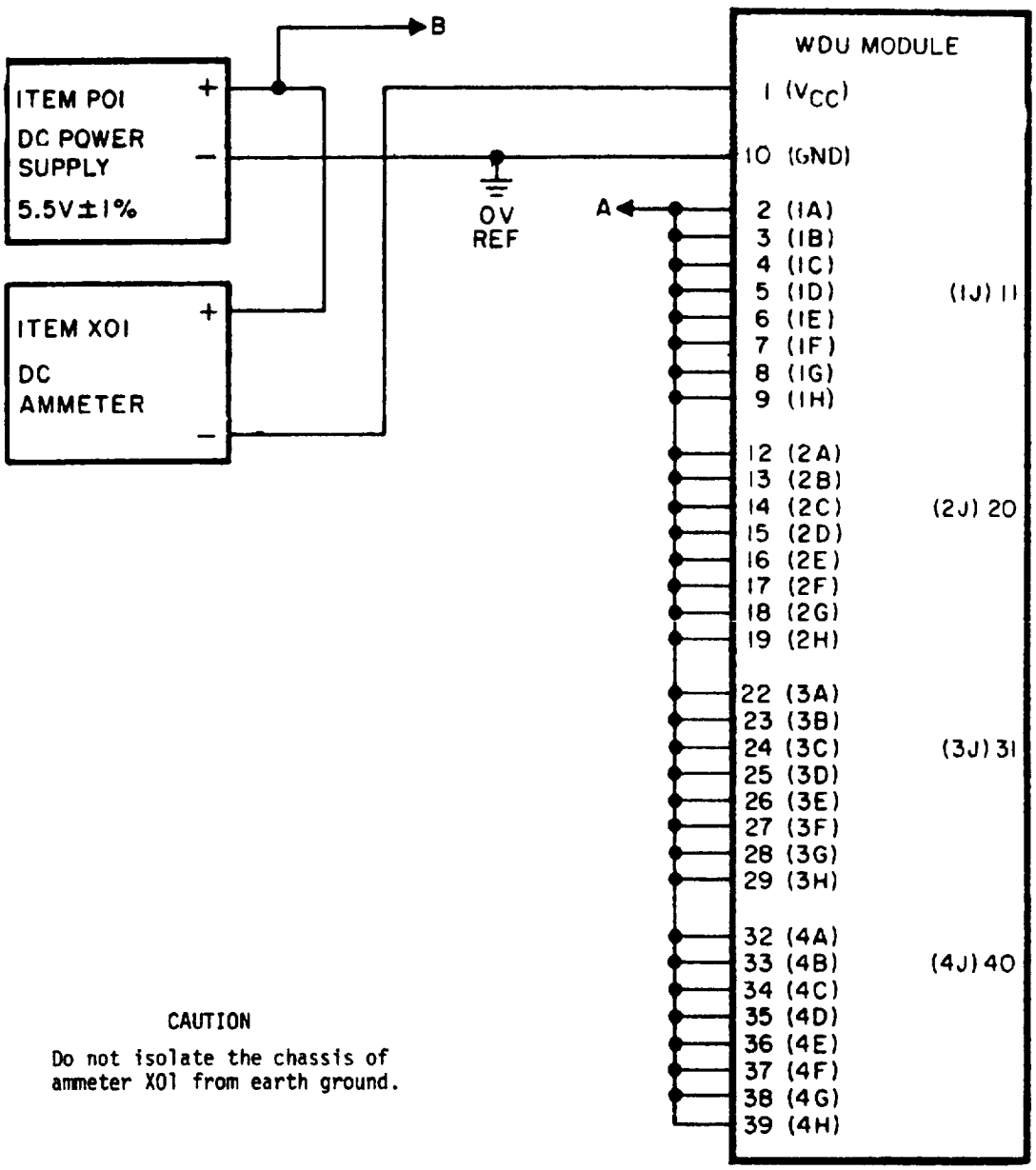


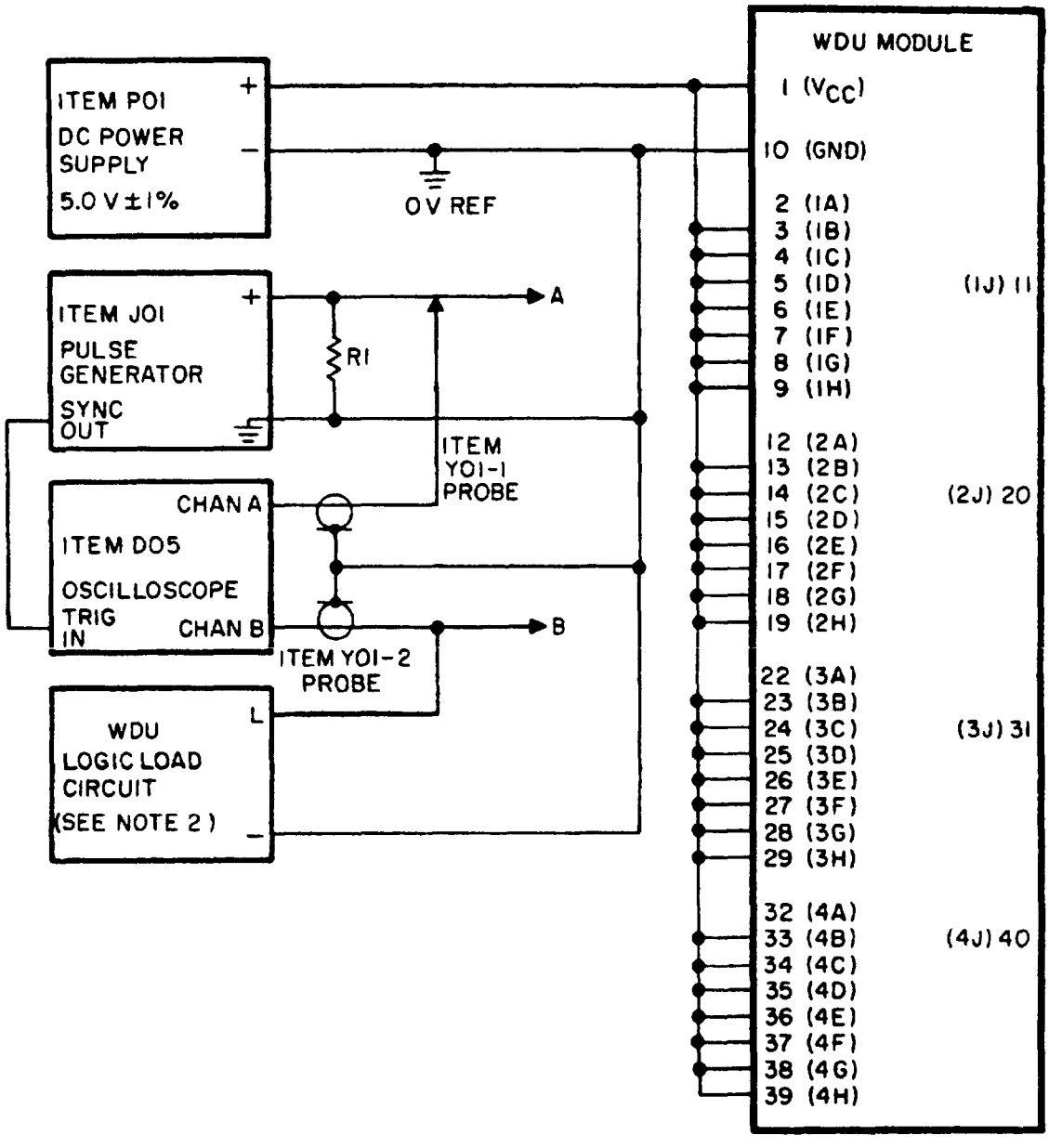
FIGURE 11. Power supply current (low-level and high-level) test circuit for table VIII.

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* TABLE IX. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Transition time, t_{THL}	2	11	3	20	ns	12,13
K2	↓	12	20	↓	↓	↓	↓
K3	↓	22	31	↓	↓	↓	↓
K4	↓	32	40	↓	↓	↓	↓
L1	Transition time, t_{TLH}	2	11	↓	↓	↓	↓
L2	↓	12	20	↓	↓	↓	↓
L3	↓	22	31	↓	↓	↓	↓
L4	↓	32	40	↓	↓	↓	↓
M1	Propagation delay time, t_{PLH}	2	11	↓	↓	↓	↓
M2	↓	12	20	↓	↓	↓	↓
M3	↓	22	31	↓	↓	↓	↓
M4	↓	32	40	↓	↓	↓	↓
N1	Propagation delay time, t_{PHL}	2	11	↓	↓	↓	↓
N2	↓	12	20	↓	↓	↓	↓
N3	↓	22	31	↓	↓	↓	↓
N4	↓	32	40	↓	↓	↓	↓

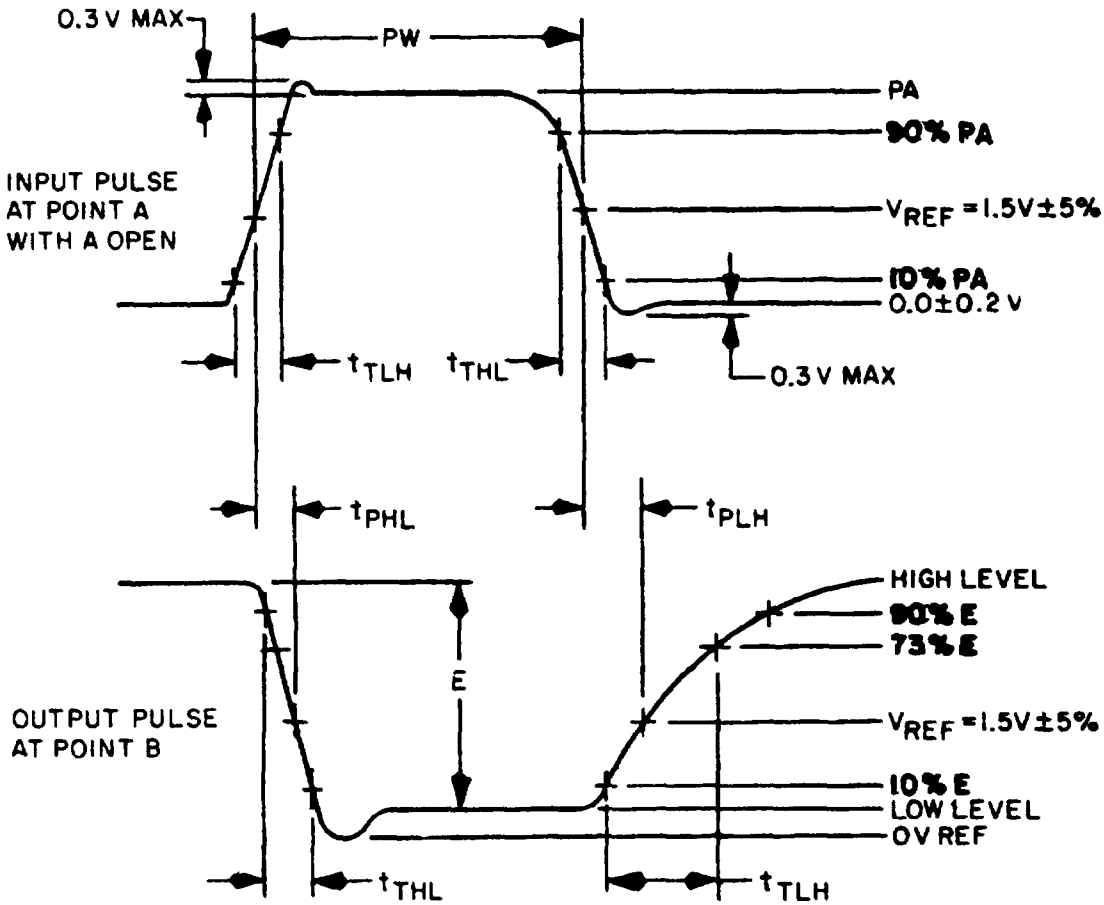
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NOTES.
1. R1 = 49.9Ω ±1%, 1 W.
2 Load each output with a load as specified on figure 9.
3. Verify that all outputs have a low level ≤ 0.45 V and a high level ≥ 2.4 V before making any measurements.

* FIGURE 12. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test cirucit for table IX.

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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 1.0 MHz ±10%.

* FIGURE 13. Transition time (t_{THL} and t_{TLH}) and propagation delay time, t_{PLH} and t_{PHL}) test circuit waveforms for table IX.

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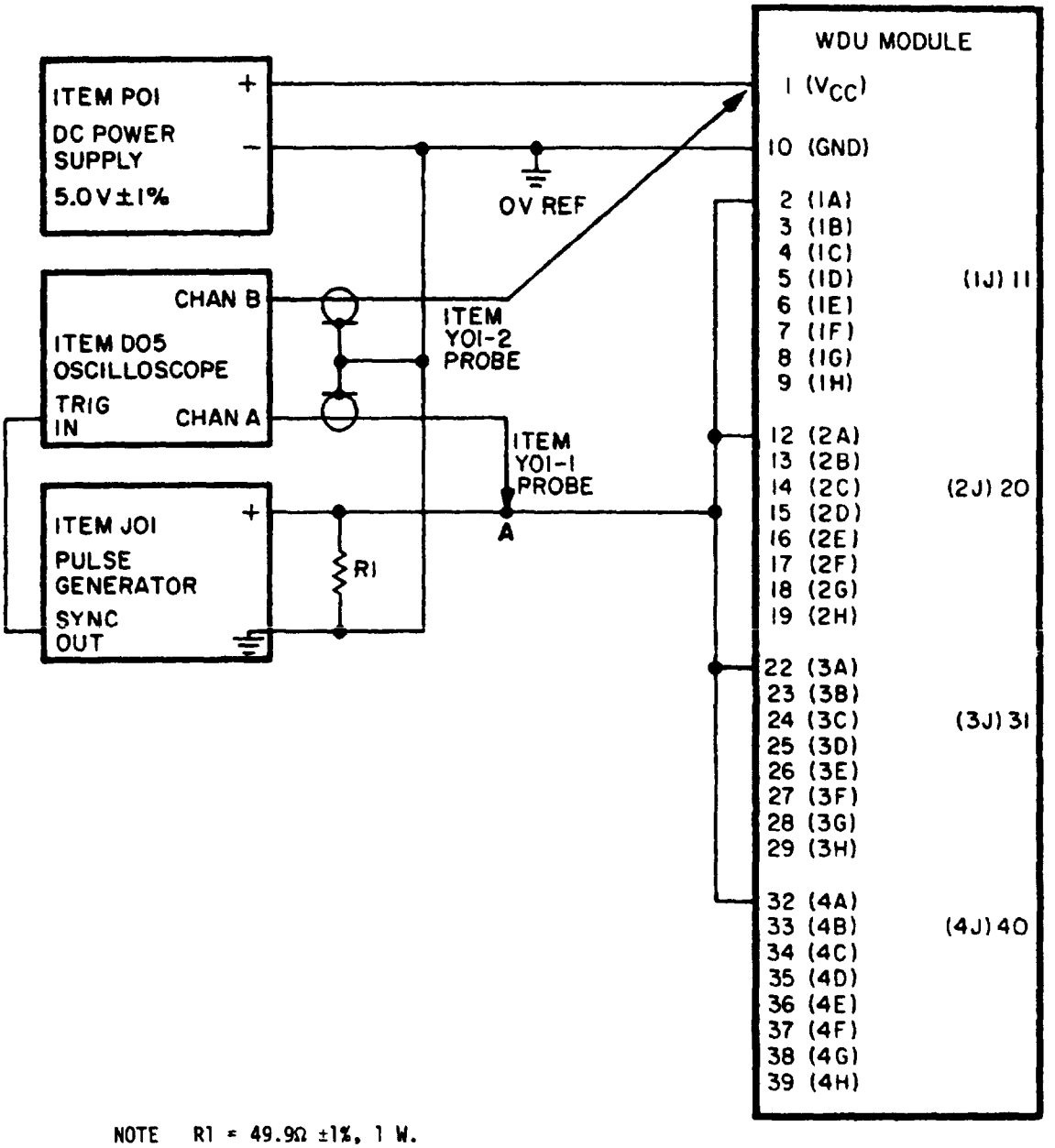
TABLE X. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient amplitude	0	400	mV	14,15

TABLE XI. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Isolation	C	Frame	10	∞	MΩ	16

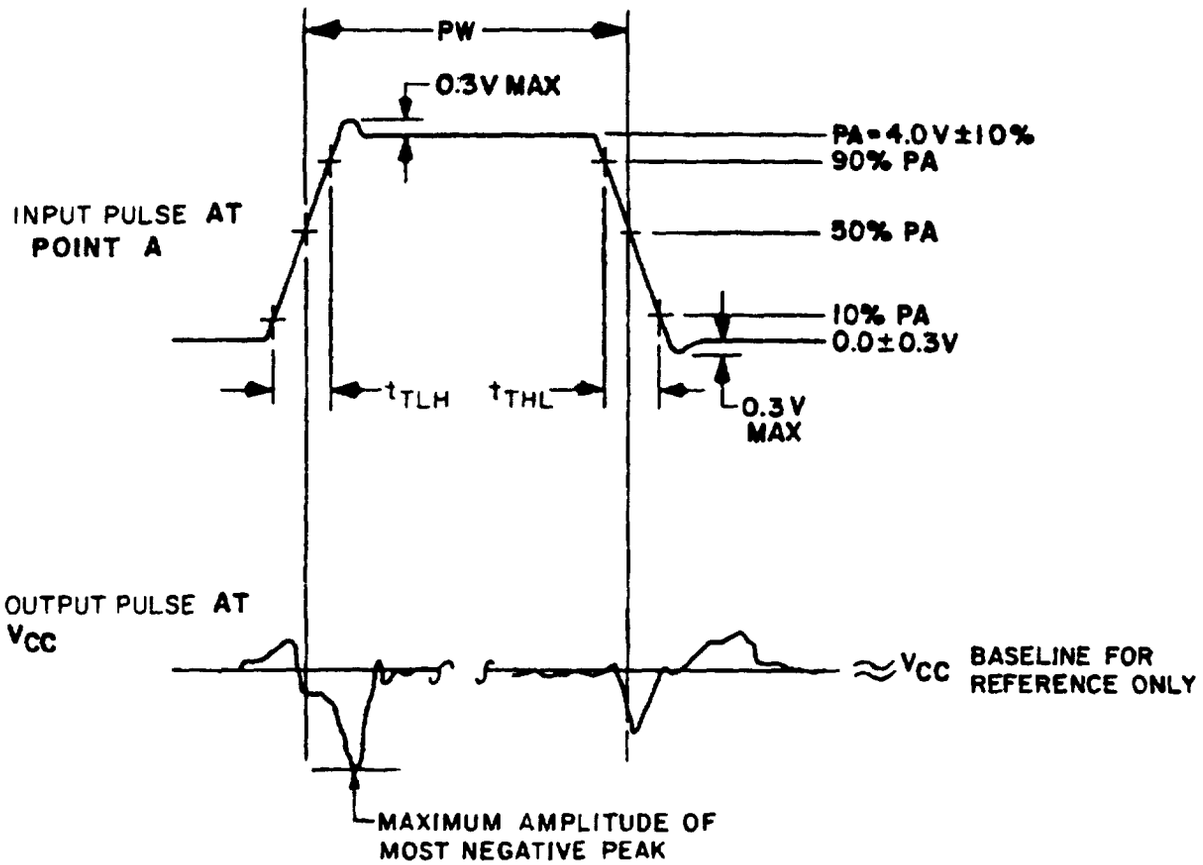
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NOTE R1 = 49.9Ω ±1%, 1 W.

FIGURE 14. Power supply transient amplitude test circuit for table X.

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NOTE: Input pulse parameters:
PW = 3 µs ±10%.
$t_{TLH} = t_{THL}$ = 10 ns ±10%.
PRR = 100 kHz ±10%.

* FIGURE 15 Power supply transient amplitude test circuit waveforms for table X.

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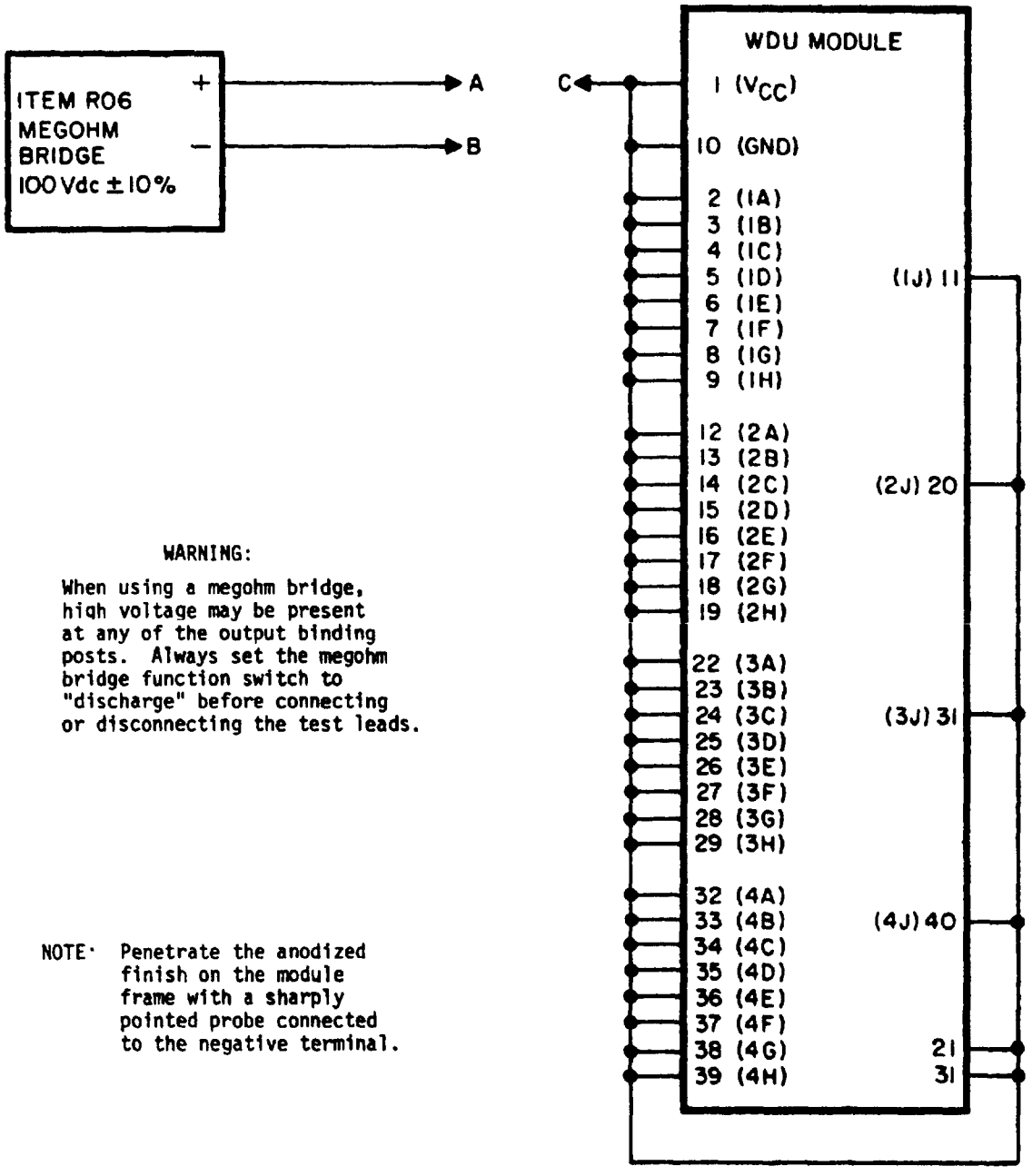


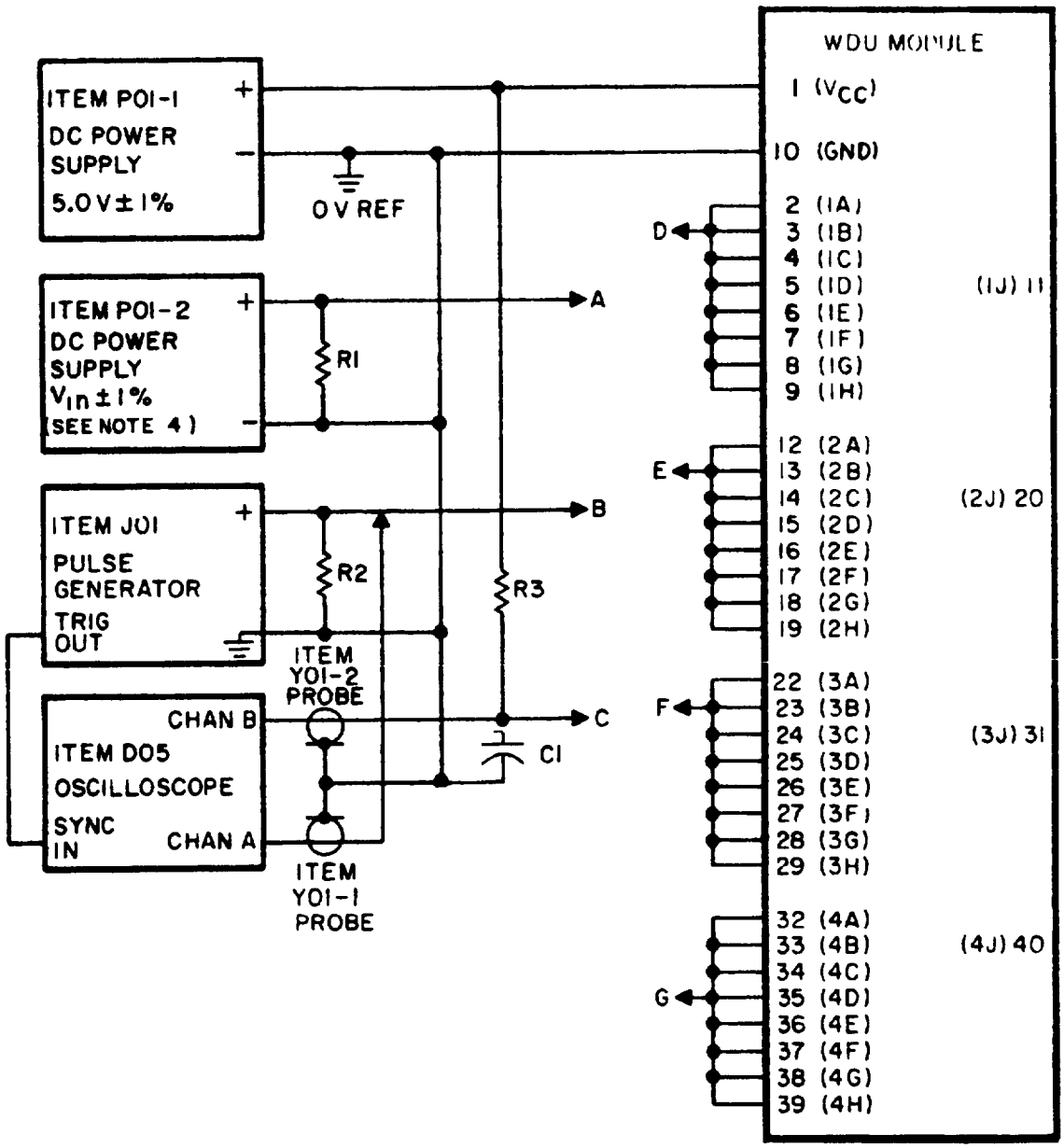
FIGURE 16. Isolation test circuit for table XI.

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* TABLE XII. Output crosstalk test.

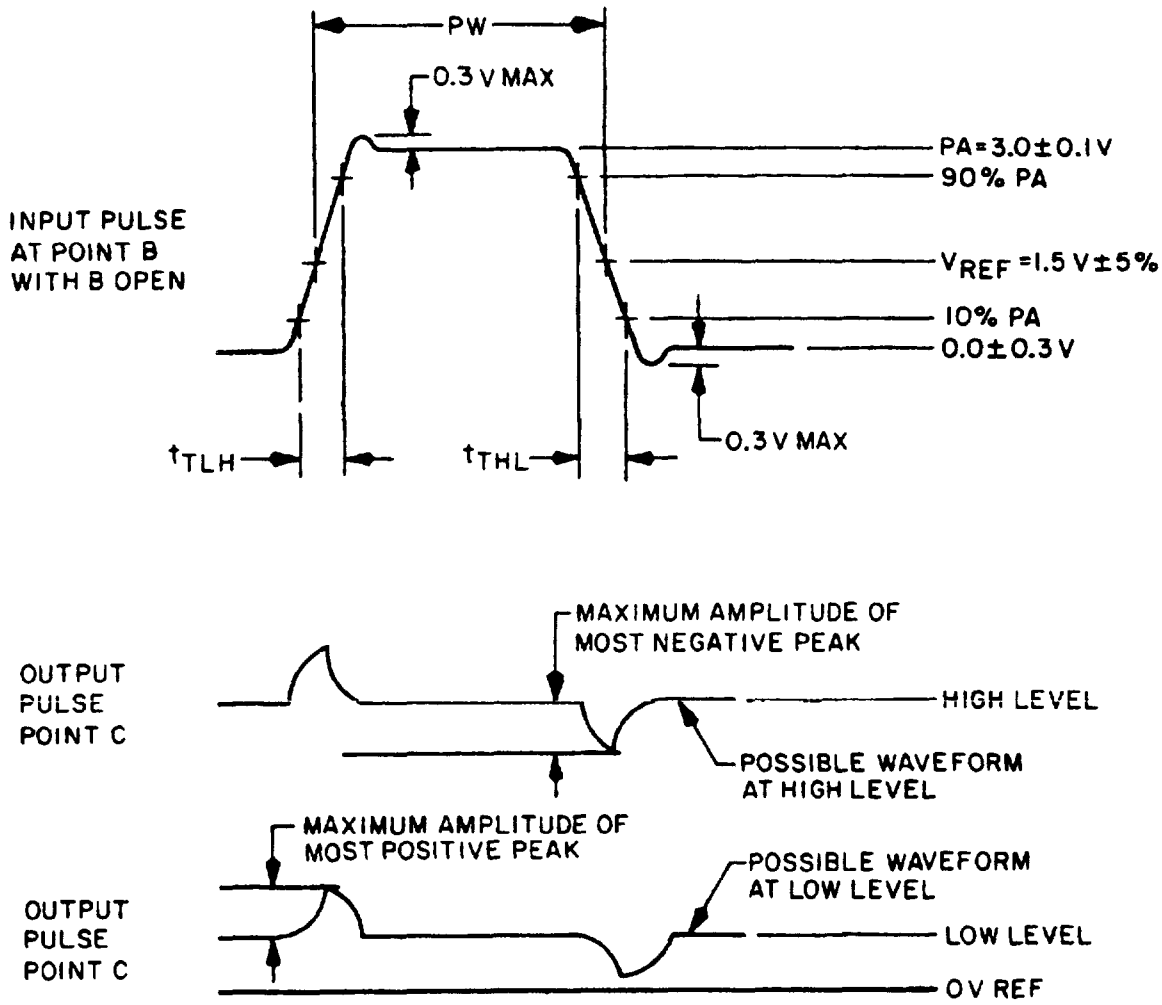
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1J Output crosstalk, low level	D	E,F,G	11	0	400	mV	17,18
R2	2J	E	D,F,G	20				
R3	3J	F	D,E,G	31				
R4	4J	G	D,E,F	40				
S1	1J Output crosstalk, high level	D	E,F,G	11				
S2	2J	E	D,F,G	20				
S3	3J	F	D,E,G	31				
S4	4J	G	D,E,F	40				



NOTES:
1. R1 = R2 = 49.9Ω ±1%, 1 W.
2. R3 = 243Ω ±1%, 1/4 W.
3. C1 = 30 pF ±1% (including probe and parasitic capacitance).
4. Set item P01-2 voltage as follows:
   V_{IN} = 2.4 V ±1% for test conditions R1-R4.
   V_{IN} = 0.4 V ±1% for test conditions S1-S4.
5. Only the output under test shall be loaded. The loop length from pin 1 to each load resistor and output, including the length of the resistor, shall not exceed 4.5 inches (114.3 mm).

* FIGURE 17. Output crosstalk test circuit for table XII.

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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 25 ±5 ns.
PRR = 1.0 MHz ±10%.

FIGURE 18. Output crosstalk test circuit waveforms for table XII.

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TABLE XIII. Input capacitance test.

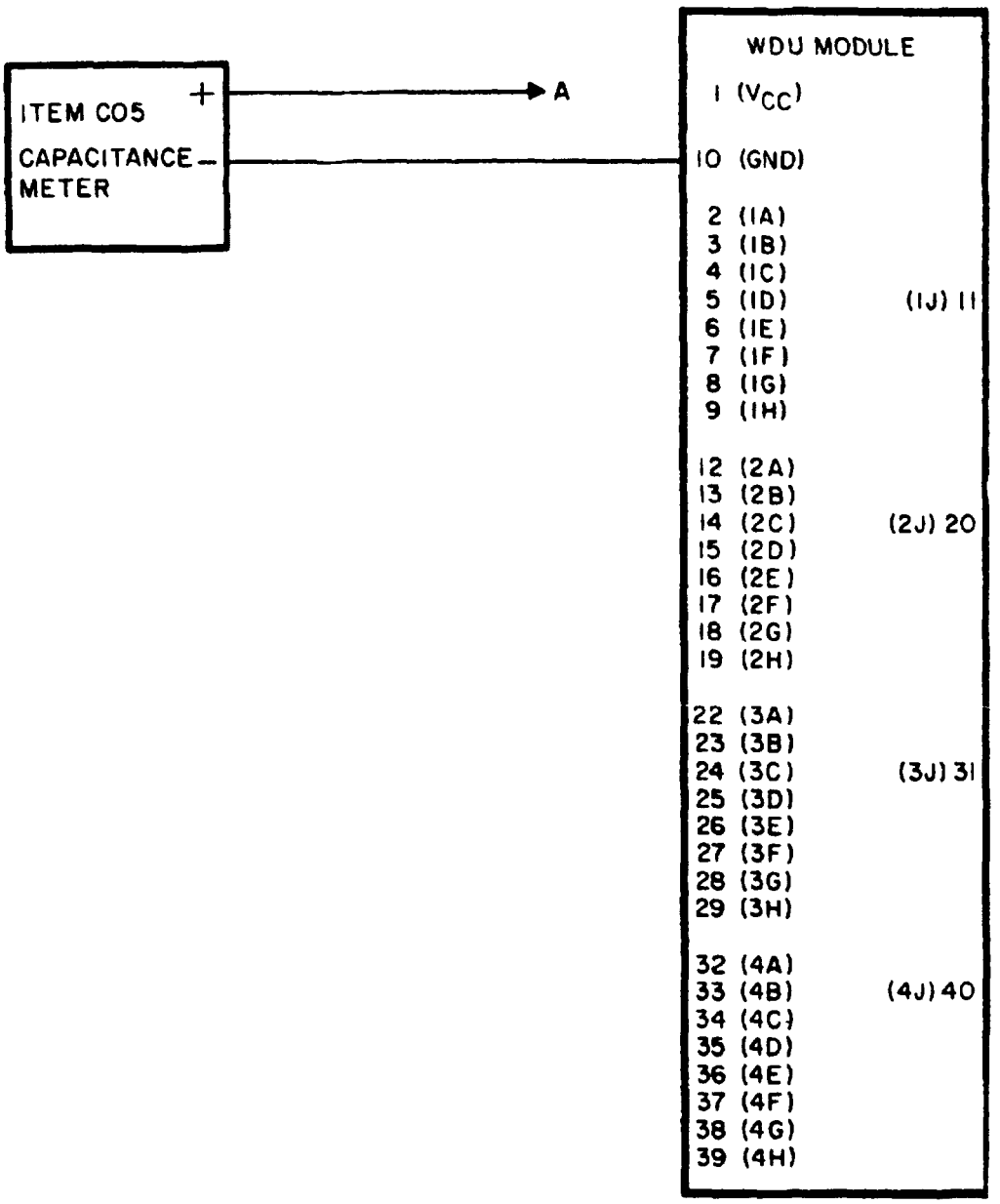
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	1A Input capacitance	2	2	12	pF	19
T2	1B	3				
T3	1C	4				
T4	1D	5				
T5	1E	6				
T6	1F	7				
T7	1G	8				
T8	1H	9				
T9	2A	12				
T10	2B	13				
T11	2C	14				
T12	2D	15				
T13	2E	16				
T14	2F	17				
T15	2G	18				
T16	2H	19				
T17	3A	22				
T18	3B	23				
T19	3C	24				
T20	3D	25				
T21	3E	26				
T22	3F	27				

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TABLE XIII. Input capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T23	3G	Input capacitance	28	2	12	pF	19
T24	3H	↓	29	↓	↓	↓	↓
T25	4A		32				
T26	4B		33				
T27	4C		34				
T28	4D		35				
T29	4E		36				
T30	4F		37				
T31	4G		38				
T32	4H		39				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting to the module.

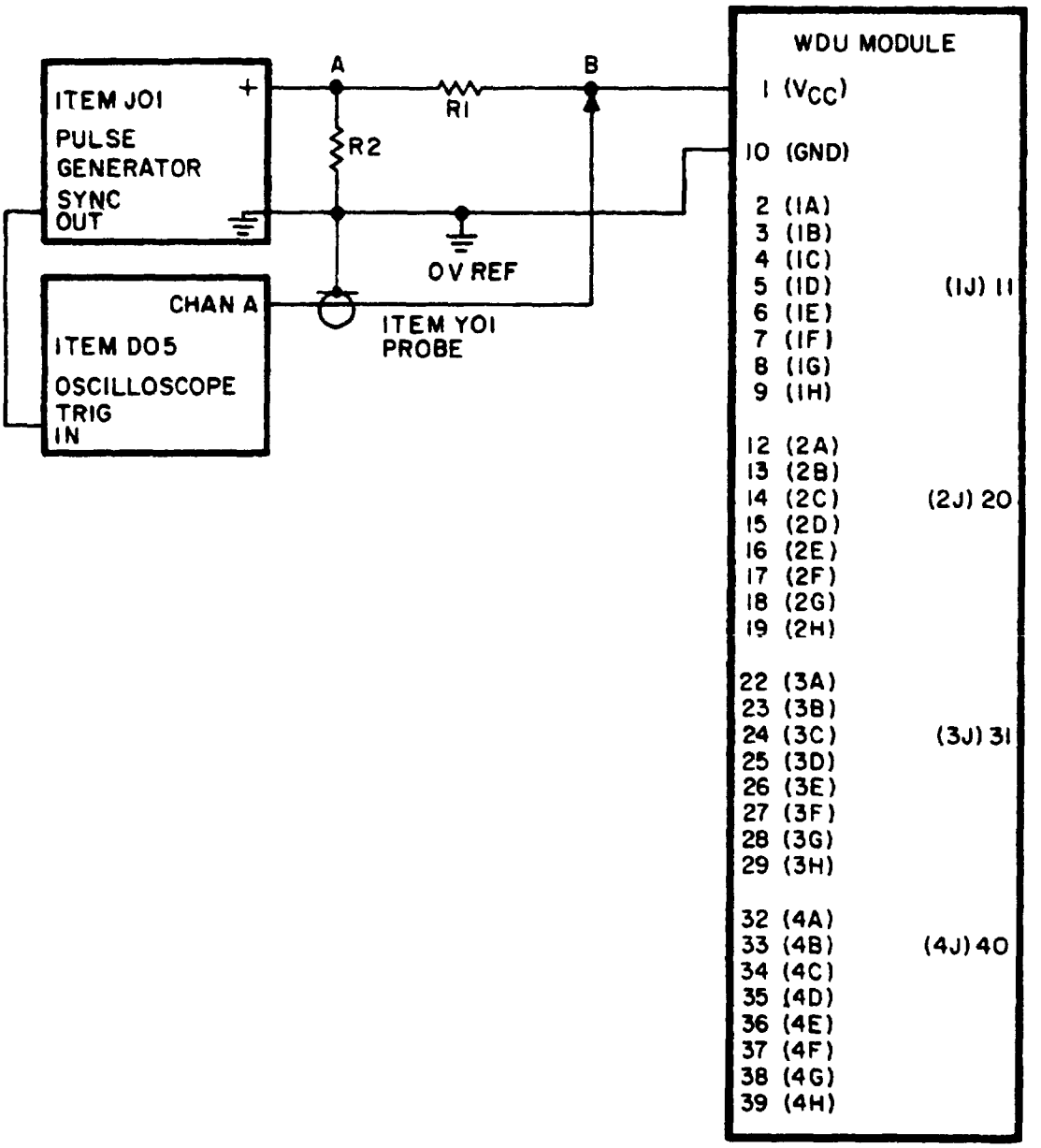
FIGURE 19 Input capacitance test circuit for table XIII.

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TABLE XIV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient rise time	2.5	7.5	µs	20,21

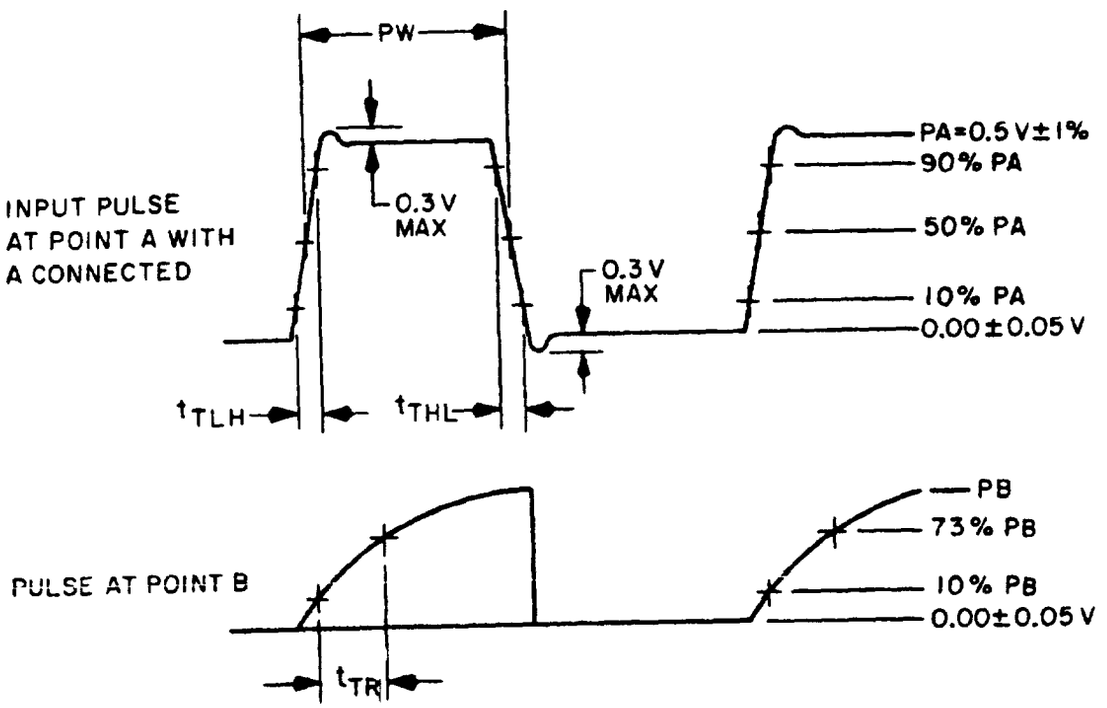
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 49 9Ω ±1%, 1 W.

FIGURE 20. Power supply transient rise time test circuit for table XIV

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure transient rise time (t_{TR}) at point B.

* FIGURE 21 Power supply transient rise time test circuit waveforms for table XIV.

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This amendment forms a part of MIL-M-28787/21B(EC), dated 22 June 1982, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. Test conditions A, B, C, H, and Q are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, H, or Q are part of a sequence of tests, the order of testing shall be as follows: Q, A, B, C, H where applicable, followed by the remaining tests in the sequence."

4.5.1 delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test condition A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through G1, and J through P, R, S and U shall be performed in the order specified in 4.4.2, using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

4.5.2, delete and substitute the following:

"4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U shall be performed using the initial limits at 25°C ±5°C ambient."

4.5.3, delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through G1, and J through P and R through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through G1, and J through P and R through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I."

PAGE 22

FIGURE 9, ITEM P01, POWER SUPPLY block: Add "+" and "-".

PAGE 41

FIGURE 13, INPUT PULSE WAVEFORM: Delete "PA" and substitute "PA = 3.0 ±0.1V".

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FIGURE 17, ITEM P01-2: Delete "V_{in}" and substitute "V_{IN}".

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